Kirti Pratihar

Email: pratiharkirti@gmail.com | Phone: (+91) 7044166375 | LinkedIn: linkedin.com/in/kirti-pratihar
GitHub: github.com/KirtiPratihar

Summary

Aspiring AI/ML engineer and B.Tech Computer Science student specializing in Artificial Intelligence. Experienced in building scalable Generative AI systems (RAG pipelines), optimizing machine learning models, and contributing to open-source software. A disciplined problem solver with a strong foundation in Data Structures Algorithms, eager to apply deep learning skills in an enterprise environment.

Technical Skills

- **Programming Languages**: Python, C++, Java, SQL
- **AI Machine Learning**: LLMs, RAG Architectures, Vector Databases (Pinecone), Neural Networks, TensorFlow, PyTorch, Scikit-learn
- **Cloud Tools**: AWS, Microsoft Azure (DP-900), Git, GitHub, Docker, Postman
- **Data Science**: Pandas, NumPy, Matplotlib, Seaborn, PySpark

Work Experience

Technical Lead, Matrix Club, *VIT Bhopal University*
Jan 2025 - Present

- Led technical operations for the 'Escape the Matrix' event, managing the digital registration infrastructure for 50+ participating teams.
- Served as the primary technical point of contact, ensuring seamless digital communication and support for club members.

Open Source Contributor (Supercontributor), *Hacktoberfest*
Oct 2025

- Achieved **'Supercontributor'** status for high-impact contributions to the open-source community.
- Refactored core Python modules and implemented cross-platform clipboard functionality, significantly improving library stability.
- Designed and integrated comprehensive pull request templates, standardizing the developer workflow for future contributors.

Technical Training, *India Space Lab Winter Internship*
Dec 2024 - Jan 2025

- Executed advanced drone flight operations and analyzed space technologies including CubeSat integration and CanSat deployment.

Projects

Generative AI RAG Chat Assistant (YAP ENGINE) GitHub Link

- Architected a Retrieval-Augmented Generation (RAG) system utilizing **Vector Search** to provide context-aware answers.
- Integrated **Pinecone** for semantic indexing and the **Groq API** for high-performance LLM inference.
- Engineered the response pipeline to reduce AI hallucinations by anchoring LLM outputs in retrieved vector data.

Credit Card Approval Prediction System GitHub Link

- Developed a predictive model for credit card approvals using a Random Forest Classifier with **90% accuracy**.
- Implemented an end-to-end ML pipeline including SMOTE for class balancing and automated feature engineering.

Taxi Trip Duration Prediction Analysis GitHub Link

- Optimized an XGBoost model to predict taxi trip durations, achieving a Root Mean Squared Log Error (RMSLE) of **0.3151**.
- Performed advanced feature engineering using K-Means clustering on geographic and temporal data.

Breast Cancer Prediction Using ML Models GitHub Link

- Engineered a comparative analysis pipeline for Logistic Regression, Decision Tree, and SVM models.
- Achieved a perfect **1.00 F1-Score** and **100% accuracy** on the test dataset using the optimized SVM model.

Education

VIT Bhopal University, *B.Tech in Computer Science (AI & Machine Learning) | CGPA: 8.71*
Bhopal, India — *Expected 2027*

Kendriya Vidyalaya Bamangachi, *XII (CBSE) - 84.4%*
— 2023

Achievements & Certifications

- **Hacktoberfest 2025 Supercontributor**: Recognized for exceptional contributions to open-source software.
- **LeetCode Profile**: Solved 500+ Data Structures & Algorithms problems; maintained 500-day consistency streak.
- Awarded 1st Prize at InnovMinds Expo Hackathon (Feb 2025) for a dementia support app utilizing AI classification.
- Ranked 79 in India Space Lab Internship Exam (Merit-based national competition).
- **Microsoft Certified**: Azure Data Fundamentals (DP-900).
- Advanced Learning Algorithms - Stanford University (May 2025).
- Introduction to TensorFlow for AI, Machine Learning, and Deep Learning - DeepLearning.AI (March 2025).